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Education

University of Waterloo, Waterloo, ON Sep 2019 – Sep 2025
Ph.D. in Computer Science – Advisor: Chengnian Sun (GPA: 90.75/100)

Northeastern University, Boston, MA Sep 2016 – Dec 2018
B.S. in Computer Science – Minor in Computer Engineering (GPA: 3.9/4.0)

Experience (Selected)

Visiting Researcher, *University of Waterloo* Sep 2025 – Present
Research Assistant, *University of Waterloo* Sep 2019 – Sep 2025

Evaluating LLMs for Type Inference on Java Code Snippets 2024 – Present

- Designed *novel evaluation* framework for LLM-based type inference on code snippets.
- Identified critical *data leakage* issue in prior evaluations of LLM-based type inference.

Scalable Type Inference on Java Code Snippets 2022 – Present

- Designed iterative optimizing algorithm to improve type inference scalability over large databases.
- Optimized SQL database queries with tables containing *millions* of Java types and methods.
- Improved running time performance by over 20x, while maintaining high accuracy.

Automated Type Inference on Java Code Snippets 2020 – 2022

- Developed a novel type inference method, leveraging semantic information in code structures.
- Engineered static analysis tool using Eclipse JDT to extract relevant type information.
- Published SnR, *doubling* performance over prior tools by *guaranteeing* type-compatible solution.

Bash Empirical Study 2019 – 2021

- Published large-scale study by scraping and processing >1 million Bash scripts from GitHub.
- Examined the common usage, code smell, and bugs in Bash scripts.
- Conducted manual analysis of bug reports to classify common bug types.

Teaching Assistant, *University of Waterloo* Sep 2019 – Aug 2025

- Held sprint meetings to verify project progression and to ensure stakeholder goals are met.
- Provided feedback to students through office hours, group meetings, and markings.
- Courses: Sequential Programs (CS241), App Dev (CS346), Software Design (CS446), Software Testing (SE465), Object Oriented Programming (CS246), Functional Programming (CS135)

Technical Knowledge

Languages

- Java, Python, JavaScript, MySQL, C, C++, Bash

Frameworks

- ANTLR, Spring, Hibernate, React, jQuery, node.js, JSP, JUnit, Maven

Publication (Selected)

Scitix: Scalable Constraint-Based Type Inference for Code Snippets with Unknown Types. * Revision
Evaluating the Genuine Type Inference Capabilities of LLMs for Java Code Snippets. * Revision
SnR: Constraint-Based Type Inference for Incomplete Java Code Snippets. * ICSE 2022
Bash in the Wild: Language Usage, Code Smells, and Bugs. * TOSEM 2022
LPR: Large Language Models-Aided Program Reduction. ISSTA 2024
Revisiting the Evaluation of Deep Learning-Based Compiler Testing. IJCAI 2023